

TS Academy Capstone Project Report

Driver Performance and Safety Analysis

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1. Introduction

This report contains the findings from my data analysis capstone project. The project focused on analysing a logistics operations database to gain meaningful insights into driver performance and safety.

The analysis was carried out using Microsoft Excel. Pivot tables were used to summarise the data, while an interactive dashboard was created to visualise the insights clearly.

The logistics industry relies heavily on drivers for smooth operations and service delivery. Understanding which drivers generate the most revenue, maintain better delivery performance, and contribute to safety is very important for decision-making and operational efficiency.

The aim of this report is to provide clear, data-driven insights that can help improve performance and safety within the company.

2. Dataset Overview

The dataset used for this project is the Logistic Operations Database, which contains 14 different sheets covering several areas of a logistics company's operations.

For this analysis, three datasets were selected because they provide a complete view of driver performance and safety.

Dataset	Description
drivers.csv	Contains driver details such as name, years of experience, home terminal, and employment status
driver_monthly_metrics.csv	Contains monthly driver performance data including revenue, completed trips, on-time delivery rate, MPG, and idle hours
safety_incidents.csv	Contains records of safety incidents including incident type, fault status, preventability, and financial claims

These datasets were selected because together they help explain driver performance from different perspectives revenue generation, punctuality, fuel efficiency, and safety records.

Driver IDs were used throughout the analysis instead of names to avoid confusion since multiple drivers may have similar names.

3. Methodology

The process followed for this analysis is as follows:

- i. The datasets were imported into Microsoft Excel, with each dataset placed on a separate worksheet
- ii. The month column in the driver_monthly_metrics dataset was formatted as a date field to allow proper time-based filtering
- iii. Five pivot tables were created to summarise important performance metrics across drivers
- iv. PivotCharts were generated from the pivot tables and moved into a dedicated Dashboard sheet
- v. A scatter plot was manually created by combining years_experience from the drivers dataset with average on-time delivery rates calculated using AVERAGEIF formulas
- vi. Four interactive slicers were added to make the dashboard dynamic and easier to filter by month, trips completed, preventable incidents, and at-fault incidents

4. Key Findings

4.1 Top 10 Drivers by Revenue

Revenue was aggregated across all months using a pivot table that summed total revenue for each driver. The top 10 revenue-generating drivers were then visualised using a column chart.

1. The highest-earning driver, **DRV00051**, generated approximately **\$2.31 million** in total revenue during the analysis period
2. The revenue gap between the highest and lowest earners within the top 10 drivers was approximately **\$97,000**, showing that revenue distribution among top drivers is competitive but not equal

4.2 Top 10 Drivers by On-Time Delivery Rate

Average on-time delivery rates were calculated for each driver across all months to evaluate punctuality across the fleet. The analysis showed that the most punctual driver, **DRV00120**, achieved an average on-time delivery rate of 50.7%, the highest recorded in the dataset.

However, the overall average on-time delivery rate across all drivers was approximately **44.6%**, which is relatively low. This suggests that, on average, more than half of all deliveries are delayed across the company.

The findings indicate that the issue may not be caused solely by individual driver performance. Other operational factors, such as route planning, scheduling systems, traffic conditions, and delivery coordination, may also be contributing significantly to low punctuality rates.

4.3 Top 10 Drivers by Fuel Efficiency (MPG)

Fuel efficiency is an important operational metric in the logistics industry because lower MPG leads to higher fuel expenses per mile travelled. Improving fuel efficiency can therefore help reduce operational costs significantly.

The analysis showed that **DRV00105** ranked as the most fuel-efficient driver overall.

The analysis also revealed little overlap between the most fuel-efficient drivers and the highest revenue-generating drivers. This suggests that productivity and fuel efficiency may operate independently, meaning drivers who generate the most revenue are not always the ones using fuel most efficiently.

4.4 Driver Experience vs On-Time Delivery Rate

A scatter plot was created to examine the relationship between years of driving experience and on-time delivery performance. Each point on the chart represented a driver, with years of experience plotted against average on-time delivery rate.

The analysis showed no strong relationship between experience and punctuality. Drivers with only a few years of experience often performed at similar on-time delivery rates as drivers with more than 20 years of experience.

This finding suggests that experience alone is not a reliable indicator of delivery performance within the company.

4.5 Incidents by Type

The analysis carried out revealed that DOT Violations were the most common incident type, accounting for nearly 23% of all recorded incidents. These violations are compliance-related issues that can expose the company to penalties, audits, and operational disruptions.

Out of the 170 incidents recorded, only 54 were classified as being directly caused by drivers. This means that approximately 68% of incidents were influenced by external or environmental conditions rather than driver fault alone.

Additionally, 82 incidents were marked as preventable, suggesting that improvements in driver training, safety awareness, vehicle maintenance, and operational procedures could help reduce future incident occurrences.

5. Recommendations

Based on the findings from this analysis, the following are some recommendations suggested for the company:

5.1 Improve On-Time Delivery Performance

Since the average on-time delivery rate is only **44.6%**, the company should investigate broader operational issues such as traffic patterns, route planning, and scheduling conflicts. Improving these areas could significantly increase customer satisfaction and operational efficiency.

5.2 Strengthen DOT Compliance Training

Since DOT Violations are the most common type of incident, the company should invest more in compliance training, vehicle inspection procedures, and safety awareness programs to reduce violations and avoid penalties.

5.3 Rethink Experience-Based Assumptions

The analysis showed that years of experience do not automatically guarantee better delivery performance. Because of this, performance evaluations should focus more on measurable metrics rather than seniority alone.

5.4 Reduce Preventable Incidents

With 82 preventable incidents recorded, the company should identify the major causes and introduce targeted solutions such as driver safety programs, fatigue management policies, and improved vehicle maintenance schedules.

5.5 Reward High-Performing Drivers

Drivers such as **DRV00051** (highest revenue), **DRV00120** (highest punctuality), and **DRV00105** (best fuel efficiency) should be recognised and used as performance benchmarks. Incentive programs based on these metrics could encourage better overall performance across the fleet.

6. Conclusion

This capstone project showed how data analysis can be used to better understand driver performance and safety within a logistics company. By analysing driver profiles, monthly performance data, and safety incidents, several valuable insights were identified.

The analysis revealed that:

- Top revenue-generating drivers are not necessarily the most punctual
- Driver experience does not strongly influence on-time delivery performance
- DOT Violations are the company's most significant safety concern

These findings highlight the importance of data-driven decision-making in logistics and fleet management.

The interactive Excel dashboard created for this project allows users to explore the data dynamically using filters such as month, trips completed, and safety indicators. This makes the analysis more accessible and easier to use for operational decision-making.

Overall, the project provided a detailed and actionable assessment of driver performance and safety, while also demonstrating how business intelligence tools can support operational improvement within the logistics industry.